UDEMY INSTRUCTOR STANDARD SCREENCAST & CODING ARCHITECTURE

Engineering Crystal-Clear Audio for Udemy Courses

A production-grade roadmap covering platform compliance, acoustic treatment, dual-channel master output, and vocal clarity.

The Udemy Quality Review Process: Unlike open-upload platforms, Udemy reviews course submissions through human and automated Quality Review Teams (QRT). The #1 reason technical courses are rejected or flagged is audio: *mono audio routing to one ear, high baseline background hiss, muffled speech, or uneven loudness between lectures*.

Pillar 1: Mandatory Udemy Technical Audio Specs

SPECIFICATION	UDEMY REQUIREMENT	TARGET METRIC / PRACTICAL FIX
Channel Routing	Dual Channel / Stereo mandatory	Ensure mono mic input is mapped to **Both L & R channels**. (One-ear audio triggers immediate course rejection).
Noise Floor	No distracting background hiss/hum	Target **< -55 dB to -60 dB** quiet room level during gaps. Fan rumble and AC hum must be eradicated.
Loudness & Peaks	Clear, consistent, non-distorting	Master Integrated Loudness to **-16 LUFS (±1.0 LUFS)** with True Peaks capped strictly at **-1.0 dBTP**.
Audio Codec / Sync	AAC 256+ kbps or PCM / Zero drift	Export audio inside MP4 container as **AAC at 320 kbps, 48 kHz sample rate, 24-bit** to prevent transcoding artifacts.

Pillar 2: Acoustic Room Physics & Keyboard Isolation

Udemy learners listen via closed-back headphones for 20–40 hours per course. Room reflections (reverb) and loud mechanical keyboard chatter cause instant auditory fatigue.

Managing Boundary Reflections

Treat the wall *directly behind your monitor* and the wall *directly behind your chair*. Heavy acoustic blankets or dense fiberglass panels eliminate the flutter echo that sounds like speaking inside a tiled bathroom.

Mechanical Keyboard Neutralization

Swap clicky mechanical switches (Blue/Brown) for Silent Linear switches (Red/Yellow/Silent Black) with rubber desk mats. Never clamp your mic arm to the typing desk; clamp to an isolated side table or wall mount.

Pillar 3: Recommended Microphone Gear Tiers

Budget USB Rig ~$120

Røde PodMic USB or **Audio-Technica ATR2100x**. Tight dynamic cardioid capsule rejects ambient room reverberations and keyboard taps effortlessly.

Pro Creator Setup ~$320

Shure MV7X paired with **Focusrite Scarlett Solo (4th Gen)** or **Elgato Wave XLR**. Ultra-clean analog preamps, analog hardware gain controls, and warm low-end weight.

Mounting & Filtering ~$110

Elgato Wave Mic Arm LP (low-profile design prevents blocking monitor line-of-sight while typing code) plus a high-density dual mesh pop shield to eliminate plosives.

Pillar 4: Physical Microphone Geometry

Proper geometric placement eliminates 95% of plosives and mouth clicks at the physical source before digital processing.

The 4–6 Inch Distance
Maintain an exact 4 to 6 inch span between your lips and capsule. Moving closer triggers booming proximity bass; moving further makes the mic pick up room reflections and keyboard clicks.

45° Off-Axis Placement
Position the mic capsule at a 30°–45° angle pointing toward your mouth corner. Rapid blasts of air from plosive consonants ("P", "B", "T") will travel past the capsule rather than slamming into it.

Line-of-Sight Ergonomics
Ensure the microphone sits just below your chin line. You must comfortably view your code editor, IDE debug terminal, and terminal console without tilting or turning your head off-axis.

Pillar 5: Instructor Vocal Skills for Long-Form Courses

Udemy courses span many hours. Student retention drops when instructors exhibit fatigue, monotone droning, vocal fry, or constant breathing gasps.

The "Thought-Chunking" Delivery Method
Never speak continuously while frantically live-typing. Break complex explanations into 2–3 sentence logical chunks: state the intention, write the code in complete silence (or at gentle pace), then explain what happened. This makes video editing effortless.

Diaphragmatic Energy & Posture
Never record slouched over a desk. Sit upright with shoulders dropped or stand using a standing desk. Inhale from the abdomen so your voice retains rich, resonant vocal authority and doesn't run out of breath mid-sentence.

Oral Hydration & Click Neutralization
Dehydration makes saliva sticky, producing sharp, irritating mouth clicks audible in high frequencies. Drink lukewarm water every 15 minutes. Avoid coffee, tea with milk, or carbonated soda 2 hours prior to recording sessions.

Vocal Consistency Across Days
Recording a 15-hour course takes weeks. Maintain exact notes of your input gain dial, mic distance, and physical room setup. Warm up your vocal cords for 5 minutes each day to avoid jarring pitch changes between lectures.

Pillar 6: Hardware Gain Staging & OBS Setup

SETTING / PARAMETER	TARGET LEVEL	PURPOSE
Interface Analog Gain	-18 dBFS to -12 dBFS	Provides 12 dB of safety headroom. Energetic speech never hits 0 dB (digital clipping).
OBS Audio Track Routing	Track 1 (Mix), Track 2 (Mic Isolated)	Isolates raw voice on a dedicated audio track away from desktop audio sounds.
Downmix to Mono	Enabled in OBS Advanced Properties	Instantly mirrors single mic channel to both Left and Right headphone channels.
Sampling Frequency	48.0 kHz / 24-bit	Synchronizes perfectly with 30fps and 60fps video frame clocks with zero drift.

Pillar 7: Master Audio Processing Chain (DAW / NLE)

Process your vocal stem using this exact 5-step serialized plugin chain (works in Audacity, DaVinci Fairlight, Adobe Audition, or Reaper) to achieve the polished "Udemy Top Instructor" sound.

STAGE 1 • CLEANING

High-Pass Filter (Low-Cut) & Gentle Spectral De-noise

Clarity

Cut all frequencies below **80 Hz** (18 dB/oct slope) to eliminate HVAC rumbles and desk thuds. Apply subtle AI noise reduction (e.g., Reaper ReaFIR, iZotope Voice De-noise) targeting no more than **4 dB to 6 dB** reduction to avoid hollow robotic artifacts.

STAGE 2 • CORRECTION

Parametric EQ: Resonances & Intelligibility

Tonal Balance

Apply a gentle dip of **-2 dB at 300 Hz – 450 Hz** (Q=1.8) to remove boxy "nasal" chest resonance. Add a broad boost of **+1.5 dB at 3.5 kHz – 5 kHz** to make variable names, syntax calls, and technical diction cut through with crisp definition.

STAGE 3 • DE-ESSING

Dynamic Sibilance Attenuation

Ear Comfort

High-end boost can sharpen "S", "T", and "CH" sounds. Place a De-Esser centered between **6 kHz and 8 kHz** with 3 dB to 5 dB reduction to ensure long listening comfort on Apple EarPods and headphones.

STAGE 4 • DYNAMIC CONTROL

Two-Stage Gentle Compression

Even Level

Set Ratio to **3:1**, Attack to **20 ms**, and Release to **80 ms**. Aim for **3 dB to 4 dB** of gain reduction during normal delivery. This glues loud peaks and softer whispered sidebars to a unified volume level.

STAGE 5 • FINAL MASTER

Loudness Normalization & Peak Limiter

Compliance

Normalize lecture to **-16 LUFS Integrated** (ideal for educational platforms). Engage a True Peak brickwall limiter set firmly to **-1.0 dBTP** to prevent inter-sample distortion when Udemy compresses your video.

Udemy Launch Roadmap & Test Video Protocol

MILESTONE	CORE ACTION ITEMS	VALIDATION CRITERIA
Phase 1: Environment	Drape blankets/acoustic panels; reposition computer chassis; configure low-profile boom arm.	Noise floor < -55 dB with room silent.
Phase 2: Template Setup	Build FX Rack (EQ + Compressor + Limiter). Map OBS downmix to stereo on Track 2.	Audio output verifies in both L and R ears.
Phase 3: Udemy Test Video	Record a 2-minute sample coding lesson; submit to free *Udemy Test Video Review Service*.	Official Udemy QRT approval on audio quality.
Phase 4: Full Production	Batch record course modules with pre-calibrated mic positioning and voice warmups.	Uniform -16 LUFS output across all lectures.